Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\bar{x} = 3$ and $A = \sum_{i=1}^{n} (x_i - 3)^2$,
$B = \sum_{i=1}^{n} (x_i - 4)^2, C = \sum_{i=1}^{n} (x_i - 2)^2$
Statement-1: $\text{Min}(A, B, C) = C$.
Statement-2: Sum of the squares of deviations is least when taken from mean

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.2

The mean square deviation of a set of n observations $x_1, x_2, \ldots, x_n$ about a point c is defined as $\frac{1}{n} \sum_{i=1}^{n} (x_i - c)^2$
The mean square deviation about -2 and 2 are 18 and 10 respectively, then standard deviation of this set of observation is

(A) 3

(B) 2

(C) 1

(D) none of these

Q.3

If the standard deviation of the numbers $2, 3, a$ and 11 is 3.5, then which of the following is true?

(A) $3a^2 - 26a + 55 = 0$

(B) $3a^2 - 32a + 84 = 0$

(C) $3a^2 - 34a + 91 = 0$

(D) $3a^2 - 23a + 44 = 0$

Q.4

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If $\sum_{i=1}^{18}(x_i-8)=9$ and $\sum_{i=1}^{18}(x_i-8)^2=45$ then the standard deviation of $x_1, x_2 \ldots x_{18}$ is

(A) $4/9$

(B) $9/4$

(C) $3/2$

(D) none of these

Q.5

In any discrete series (when all the values are not same) the relationship between M.D. about mean and S.D. is

(A) M.D. = S.D.

(B) M.D. > S.D.

(C) M.D. < S.D.

(D) M.D. $\leq$ S.D.

Q.6

The mean of two samples of sizes 200 and 300 were found to be 25,10 respectively. Their standard deviations were 3 and 4 respectively. The variance of combined sample of size 500 is

(A) 64

(B) 65.2

(C) 67.2

(D) 64.2

Q.7

If the mean deviation of the numbers $1, 1+\mathrm{d}, 1+2\ \mathrm{d}, \ldots . 1+100d$ from their mean is 255, then d is equal to :

(A) 20.0

(B) 10.1

(C) 20.2

(D) 10.0

Q.8

If the standard deviation of the observations -5, $-4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ is $\sqrt{10}$. The standard deviation of observations $15, 16, 17, 18, 19, 20$, $21, 22, 23, 24, 25$ will be

(A) $\sqrt{10}+20$

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(B) $\sqrt{10}+10$

(C) $\sqrt{10}$

(D) None of these

Q.9

Suppose a population A has 100 observations $101, 102, \ldots\ldots\ldots\ldots, 200$ and another population B has 100 obsevrations $151, 152, \ldots\ldots\ldots\ldots\ldots\ldots 250$. If V_{A} and V_{B} represent the variances of the two populations, respectively then $\frac{V_A}{V_B}$ is

(A) 1

(B) $\frac{9}{4}$

(C) $\frac{4}{9}$

(D) $\frac{2}{3}$

Q.10

In a study of two groups, the following results were obtained:

	Group A	Group B
Sample Size	20	25
Sample mean	22	23
Sample standard deviation	10	12

Which of the following statements is correct?

(A) Group A is less variable than Group B because Group A's standard deviation is smaller.

(B) Group A is less variable than Group B because Group A's sample size is smaller.

(C) Group A is less variable than Group B because Group A's sample mean is smaller.

(D) Group A is less variable than group B because Group A's coefficient of variation is smaller.

Q.11

In an experiment with 15 observations on X, the following results were available $\Sigma x^2 = 2830, \Sigma x = 170$. On observation that was 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is

(A) 78.00

(B) 188.66

(C) 177.33

(D) 8.33

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Q.12

The standard deviation of $9, 16, 23, 30, 37, 44, 51$ is

(A) 7

(B) 9

(C) 12

(D) 14

Q.13

The mean and S.D. of the marks of 200 candidates were found to be 40 and 15 respectively. Later, it was discovered that a score of 40 was wrongly read as 50 . The correct mean and S.D. respectively are

(A) $14.98, 39.95$

(B) $39.95, 14.98$

(C) $39.95, 224.5$

(D) None of these

Q.14

A fair die is tossed 180 times, the standard deviation of the number of sixes equal to

(A) $\sqrt{30}$

(B) 5

(C) 25

(D) $\sqrt{90}$

Q.15

The first of two samples has 100 items with mean 15 and SD 3. If the whole group has 250 items with mean 15.6 and $SD = \sqrt{13.44}$ the SD of the second group is

(A) 5

(B) 4

(C) 6

(D) 3.52

Answers & Solutions

Q.1 Answer:

STATEMENT-1 is False, STATEMENT-2 is True

Solution:

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Among all the squares of the deviations, the least is when taken from mean.

This can be proved by considering,

$F(x) = \sum_{i=1}^{n} (x_i - X)^2$ **,** F **is least when** $X = \bar{x}$**.**

Q.2 Answer:

3

Solution:

Given that $18 = \frac{1}{n}\sum (x_i + 2)^2, 10 = \frac{1}{n}(x_i - 2)^2$

Subtracting, $8 = \frac{1}{n}\left(\sum (x_i + 2)^2 - \sum (x_i - 2)^2\right)$

$= \frac{8}{n}\sum x_i \Rightarrow \bar{x} = 1$

Also $18 = \frac{1}{n}\sum \left(x_i^2 + 4 + 4x_i\right)$

$= \frac{1}{n}\sum x_i^2 + 4 + 4\frac{1}{n}\sum x_i$

$= \frac{1}{n}\sum x_i^2 + 8$

So $10 = \frac{1}{n}\sum x_i^2$

$\sigma_X^2 = \frac{1}{n}\sum x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2 = 10 - 1 = 9$

Thus $\sigma_X = 3$

Q.3 Answer:

$3a^2 - 32a + 84 = 0$

Solution:

$\bar{x} = \frac{1}{4}(2 + 3 + a + 11) = 4 + \frac{1}{4}a$

Variance $= \frac{1}{4}\left(2^2 + 3^2 + a^2 + 11^2\right) - \left(4 + \frac{1}{4}a\right)^2$

$\left(\frac{7}{2}\right)^2 = \frac{1}{4}\left(134 + a^2\right) - \frac{1}{4}\left(8 + \frac{1}{2}a\right)^2$

$\Rightarrow 49 = 134 + a^2 - \left(64 + \frac{1}{4}a^2 + 8a\right)$

$\Rightarrow \frac{3}{4}a^2 - 8a + 21 = 0$

$\Rightarrow 3a^2 - 32a + 84 = 0$

Q.4 Answer:

$3/2$

Solution:

Let $d_i = x_i - 8$ **but**

$$\sigma_x^2 = \sigma_d^2 = \frac{1}{18}\sum d_i^2 - \left(\frac{1}{18}\sum d_i\right)^2$$

$$= \frac{1}{18} \times 45 - \left(\frac{9}{18}\right)^2 = \frac{5}{2} - \frac{1}{4} = \frac{9}{4}$$

Therefore $\sigma_x = 3/2.$

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Q.5 Answer:

M.D. < S.D.

Solution:

M.D. $= \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$ **and S.D.** $= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

Let $a_i = |x_i - \bar{x}|$ **so (M.D.** $)^2 = \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)^2$ **and (S.D.)** $^2 = \frac{1}{n}\sum_{i=1}^{n} a_i^2$**. By Cauchy Schwarz inequality**

$\left(\sum_{i=1}^{n} a_i\right)^2 < n\sum_{i=1}^{n} a_i^2$

$\Rightarrow \quad \left(\frac{1}{n}\sum a_i\right)^2 < \frac{1}{n}\sum a_i^2 \Rightarrow$ **M.D. < S.D.**

Q.6 Answer:

67.2

Solution:

Combined mean $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$= \frac{200\times 25 + 300\times 10}{500} = 16$

Let $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$ and

$d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

Now we know that

$\sigma^2 = \frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}$

$= \frac{200(9+81) + 300(16+36)}{500} = \frac{33600}{500} = 67.2$

Q.7 Answer:

10.1

Solution:

Mean $= \frac{101 + \mathrm{d}(1+2+3+\ldots\ldots+100)}{101}$

$= 1 + \frac{\mathrm{d}\times 100\times 101}{101\times 2} = 1 + 50\ \mathrm{d}$

$\because$ **Mean deviation from the mean** $= 255$ - $\frac{1}{101}[|1 - (1+50d)| + |(1+d) - (1+50d)| + |(1+2d) - (1+50d)| + \ldots + |(1+100d) - (1+50d)|$

$= 255$

$\Rightarrow \quad 2d[1 + 2 + 3 + \ldots + 50] = 101 \times 255$

$\Rightarrow \quad 2d \times \frac{50\times 51}{2} = 101 \times 255$

$\Rightarrow d = \frac{101\times 255}{50\times 51} = 10.1$

Q.8 Answer:

$\sqrt{10}$

Solution:

The new observations are obtained by adding 20 to each. Hence, σ **does not change.**

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Q.9 Answer:

1

Solution:

$\sigma_x^2 = \frac{\sum d_i^2}{n}$ **(Here deviations are taken from the mean). Since** A **and** B **both have 100 consecutive integers, therefore both have same standard deviation and hence the variance.**

$\frac{V_A}{V_B} = 1$ **(As** $\sum d_i^2$ **is same in both the cases)**

Q.10 Answer:

Group A is less variable than group B because Group A's coefficient of variation is smaller.

Solution:

For Group A:

Coefficient of variation

$CV_A = \frac{\text{S.D.}}{\text{Mean}} = \frac{10}{22} = 0.4545.$

For Group B:

$CV_B = \frac{12}{23} = 0.522$

$\Rightarrow$ **Group** A **is less variable.**

Q.11 Answer:

78.00

Solution:

Given $\Sigma x = 170, \Sigma x^2 = 2830$

Increase in $\Sigma x = 10$**, then** $\sum x' = 170 + 10 = 180$ **Increase in** $\Sigma x^2 = 900 - 400 = 500$**, then** $\Sigma x'^2 = 2830 + 500 = 3330$

$\therefore$ **Variance** $= \frac{1}{n}\Sigma x'^2 - \left(\frac{\Sigma x'}{n}\right)^2$

$= \frac{3330}{15} - \left(\frac{180}{15}\right)^2 = 222 - 144 = 78$

Q.12 Answer:

14

Solution:

Standard deviation $= \sigma = d\sqrt{\frac{n^2-1}{12}}$

$d =$ **size between each observation** $= 7$ $n =$ **total number of observation** $= 7$

$$\therefore \quad \sigma = 7\sqrt{\frac{(7)^2-1}{12}} = 7\sqrt{\frac{49-1}{12}}$$

$$= 7\sqrt{\frac{48}{12}} = 7 \times 2 = 14$$

Q.13 Answer:

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$39.95, 14.98$

Solution:

Corrected $\Sigma x = 40 \times 200 - 50 + 40 = 7990$

$\therefore$ **Corrected** $\bar{x} = 7990/200 = 39.95$

Incorrect $\Sigma x^2 = n\left[\sigma^2 + \bar{x}^2\right] = 200\left[15^2 + 40^2\right.$.

$= 365000$

Corrected $\Sigma x^2 = 365000 - 2500 + 1600 = 364100$

Corrected $\sigma = \sqrt{\frac{364100}{200} - (39.95)^2}$

$= \sqrt{(1820.5 - 1596)} = \sqrt{224.5} = 14.98.$

Q.14 Answer:

5

Solution:

$\mathrm{n} = 180, \mathrm{p} = \mathrm{P}($ **getting a six** $) = \frac{1}{6}$ **and** $\mathrm{q} = \frac{5}{6}$

So, S.D. $= \sqrt{\mathrm{npq}}$

Q.15 Answer:

4

Solution:

Use $\sigma^2 = \frac{\mathrm{n_1} \times \left(\sigma_1^2 + \mathrm{d_1^2}\right) + \mathrm{n_2} \times \left(\sigma_2^2 + \mathrm{d_2^2}\right)}{\mathrm{n_1} + \mathrm{n_2}}$

where $\mathrm{d_1} = \mathrm{m_1} - \mathrm{a}$

$\mathrm{d_2} = \mathrm{m_2} - \mathrm{a}$

$a =$ **mean of the whole group.**

Let $\mathrm{m_2} =$ **mean of the second group.**

Therefore $15.6 = \frac{100 \times 15 + 150 \times \mathrm{m_2}}{250}$

$$\Rightarrow \mathrm{m_2} = \frac{250 \times 15.6 - 100 \times 15}{150}$$

$$\Rightarrow \mathrm{m_2} = \frac{2400}{15}$$

$$\mathrm{m_2} = 16$$

$$\Rightarrow 13.44 = \frac{100 \times 9 + 150 \times \sigma^2 + 100 \times (0.6)^2 + 150 \times (0.4)^2}{250}$$

$$\Rightarrow \sigma = 4. \text{ ans}$$

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